RESEARCH PROPOSAL · T14

What we learned from our 3D affordance experiments

And the next question we plan to test



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The task is to find which points support an action

INPUT

A 3D object point cloud
about 1,000–2,000 XYZ points

✦

An action in words
“grasp”, “press”, or “sit on”

OUTPUT

A score or label for every point

The same object can have different parts highlighted for different actions.

Why it is difficult: a partial view may hide the part that matters.

Our OpenAD baseline and evaluation setup

PUBLIC CHECKPOINT · E01 ADAPTED EVALUATION

OpenAD PN2

PARTIAL · E01

12.50%

FULL · E01

14.41%

- Released weights fixed; no retraining
- E01 used an adapted evaluation; see appendix
- Separate profile: Apple M3 Pro GPU via MPS

3D-AFFORDANCENET VALIDATION

Partial views

9,140 views · 2,285 objects

E05 all-view mIoU[1]: 12.54%

Full shapes

2,285 complete objects

One 2,048-point pass: 0.546 s

~1.295 GiB sampled Metal allocation[2]

[1]E05 differs from E01 in batch/sampling setup. [2]Unified-memory sample, not peak VRAM.

E01 is an adapted baseline evaluation, not an untouched official reproduction.

OpenAD scores change sharply with query-name choice

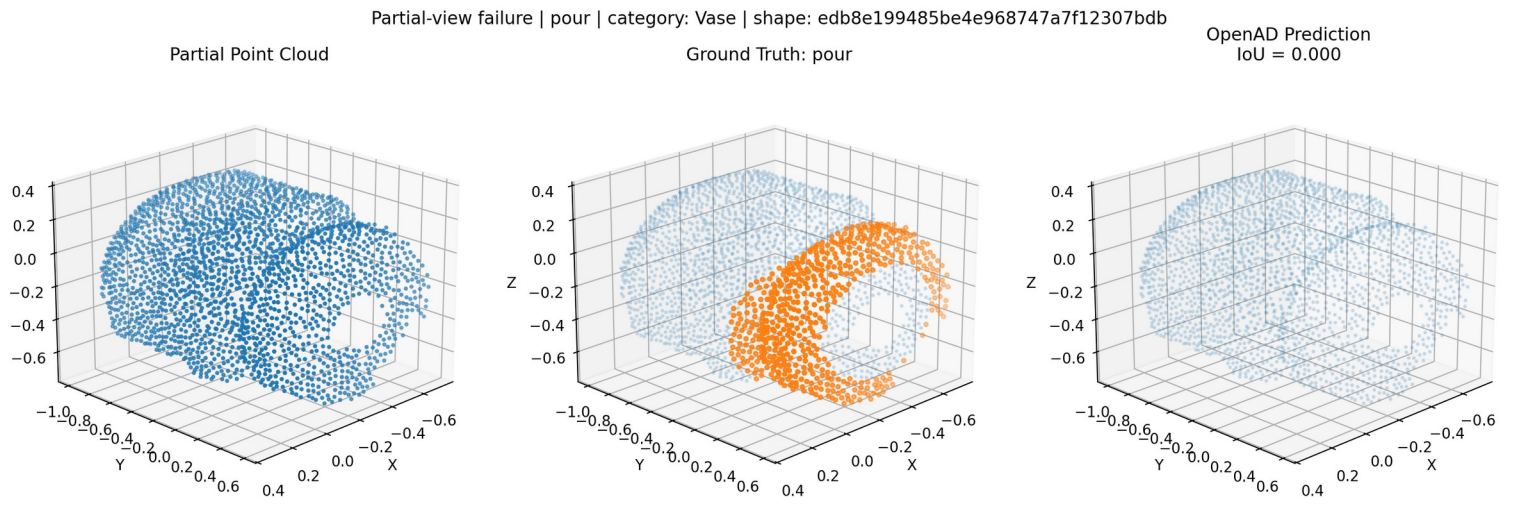
Same checkpoint and point clouds; only the query vocabulary changed.

MEAN POINT-WISE IoU[1]	TRAINING-NAME QUERIES	HELD-OUT QUERY NAMES
Partial views · 18 actions	40.39%	10.77%
Partial views · incl. “none”	41.45%	12.53%
Full shapes · 18 actions	35.18%	7.87%
Full shapes · incl. “none”	36.18%	9.76%

[1] IoU measures overlap between predicted points and the labeled part. Higher is better.

The model scored lower with held-out query names. This shows wording sensitivity, not new-concept generalization.

We found a clear failure on one partial vase



Object: vase · Query: “pour”

563

points marked “pour” in the labels

0 of 563

target points predicted correctly

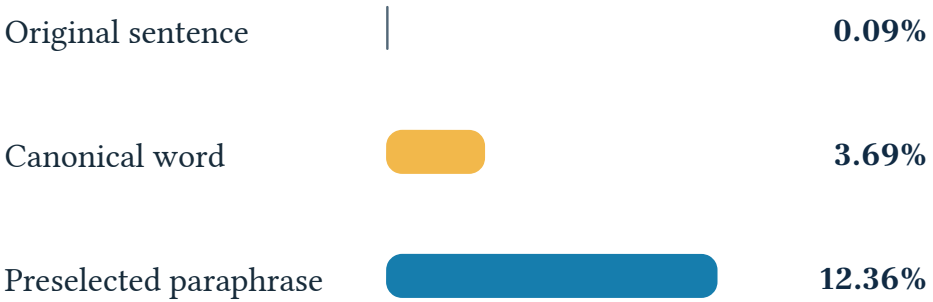
On those target points: pour score 0.058; none score 0.644.

ONE EXAMPLE — NOT AN AVERAGE

One paraphrase helped “unlock”; it did not help every query

OPENAD PROMPT-TRANSFER PILOT · 1,941 OBJECTS

“unlock” point-wise IoU



THE TRADE-OFF

Negative objects with any false alarm

0.34% → 1.61%

The other four tested prompts did not show a similar gain.

7,764 partial views · exploratory validation, not an official test set

Earlier prompt studies found mixed, action-specific effects

E08–E13 · PROMPT WORDING

Unofficial Espresso-3D + Affogato weights

No consistent prompt winner

Prompt averaging often lowered scores; effects varied by action.

E13 grab: ✦5.85 points

95% CI +1.59 to +11.00; positive objects only.

E14–E16 · PRESENCE & THRESHOLDS

At score cutoff 0.5, accommodate negative-object alarms rose 6/20 ⟶ 13/20 with the selected phrase.

A pre-set 10% alarm cap chose cutoff 1.00 and detected 0/20 positive objects.

Threshold calibration did not find a useful operating point.

Small exploratory cross-model studies; not official Affogato results. Affostruction was not run.

How do prompt strategies affect localization and false alarms?

RESEARCH QUESTION

Across 18 actions on partial point clouds, how do one paraphrase and a three-prompt average change OpenAD IoU and false alarms versus canonical action words?

TESTABLE HYPOTHESIS

Prompt effects will vary by action. We predict the three-prompt average will not give a reliable positive mean-IoU gain; any gains may come with more false alarms.

Primary contrast: three-prompt average vs canonical; report paired IoU change and false-alarm change.

A 95% CI fully above zero would overturn our prediction.

Three query conditions; everything else stays fixed

FAIR COMPARISON

- Same frozen OpenAD model and point samples
- A: canonical action word
- B: one preregistered paraphrase per action
- C: average of three fixed paraphrase score maps
- Partial views primary; full shapes control

WHAT WE WILL REPORT

- Primary: 18-action mean IoU, C vs A
- Secondary: B vs A; per-action IoU
- 19-class IoU and AUC
- False alarms on negative objects
- Paired 95% CI, resampling by object

Next experiment: compare three prompt strategies

1 · WRITE	For each of 18 actions, prepare three meaning-equivalent paraphrases. Freeze them before evaluation.
2 · RUN	Score canonical words, one preregistered paraphrase per action, and the average of three paraphrase maps on identical partial views.
3 · MEASURE	Primary: 18-action mean IoU, three-prompt average vs canonical. Also track false alarms and action-level results.
4 · CHECK	Use full shapes as a control; bootstrap paired differences by object and inspect failure cases.

Use a fresh object-disjoint test set if possible; otherwise call the study exploratory.

Week 9–15: a staged plan for the prompt comparison

WEEK 9	Freeze the research question, primary comparison, and test split.
WEEK 10	Write three paraphrases per action; freeze prompts before scoring.
WEEK 11	Test the three scoring conditions and averaging script on a small sample.
WEEK 12	Run all three query conditions on the partial-view test set.
WEEK 13	Run the full-shape control; calculate per-action IoU and false alarms.
WEEK 14	Bootstrap paired differences by object; inspect gains and failure cases.
WEEK 15	Summarize findings, limitations, and figures; finish the proposal presentation.

CHECKPOINTS

Same frozen model
Same objects and point samples
Prompt wording fixed before evaluation

If the validation set is reused, report the result as exploratory. A fresh object-disjoint test set is stronger.

Q & A

Thank you



Appendix A · baseline details and limits

CHECKPOINT & EVALUATION

OpenAD PN2 released checkpoint, frozen

SHA-256

cbcfe9ff70 448d97a7047f1ade817 645a9be29dd2782fba57 776 792a8210f2ae

Adapted run record · E03–E05 commit

b082 265ed085 455a3fdc80b0d8d95dc51 351e7b4

Four modified files: models/openad_dgcnn.py, models/openad_pn2.py, test_open_vocab.py, utils/eval.py. E01 did not record its commit.

E01: partial 12.4964% · full 14.4121%
E05: partial 12.5364% point-weighted, batch 1

APPLE M3 PRO · PYTORCH MPS

One 2,048-point forward pass: 0.5456 s after setup; no warm-up.

Sampled high-water Metal allocation: about 1.295 GiB at 5 ms sampling. Unified memory means this is not a separate VRAM peak.

E01 is an adapted evaluation, not an untouched official reproduction. E05 uses different batch and stochastic sampling conditions.

Metric values are rounded only on the main slides.

Appendix B · experiment record and sources

WHAT THE EXPERIMENTS SHOW

E01 · adapted OpenAD baseline: 12.50% partial, 14.41% full.

E02 · one partial-vase failure case.

E03–E05 · device checks and 9,140-view evaluation.

E06–E07 · smoke test and one-case Espresso inference.

E08–E13 · small prompt tests; mixed effects by action.

E14–E16 · negative alarms and threshold limits.

E17 · query-name sensitivity; E18 · prompt transfer to OpenAD.

LIMITS & REFERENCES

E08–E18 are exploratory validation results. E08–E16 use an unofficial Espresso-3D reimplementation with Affogato-pretrained weights. Affostruction inference was not run.

OpenAD · Nguyen et al., IROS 2023
arXiv:2303.02401

Affogato · Lee et al., ECCV 2026
arXiv:2506.12009

Affostruction · Park et al., CVPR 2026
arXiv:2601.09211

"Affogato results" here means local exploratory inference, not the paper's official benchmark.